UNIVERSIDAD COMPLUTENSE MADRID

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Facultad de Informática

medal

Structuring Electronic Health Records of breast cancer with Natural Language Processing

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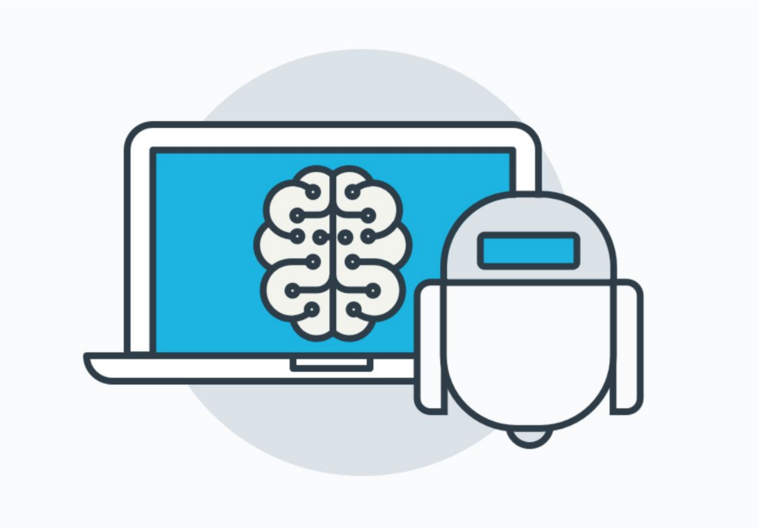
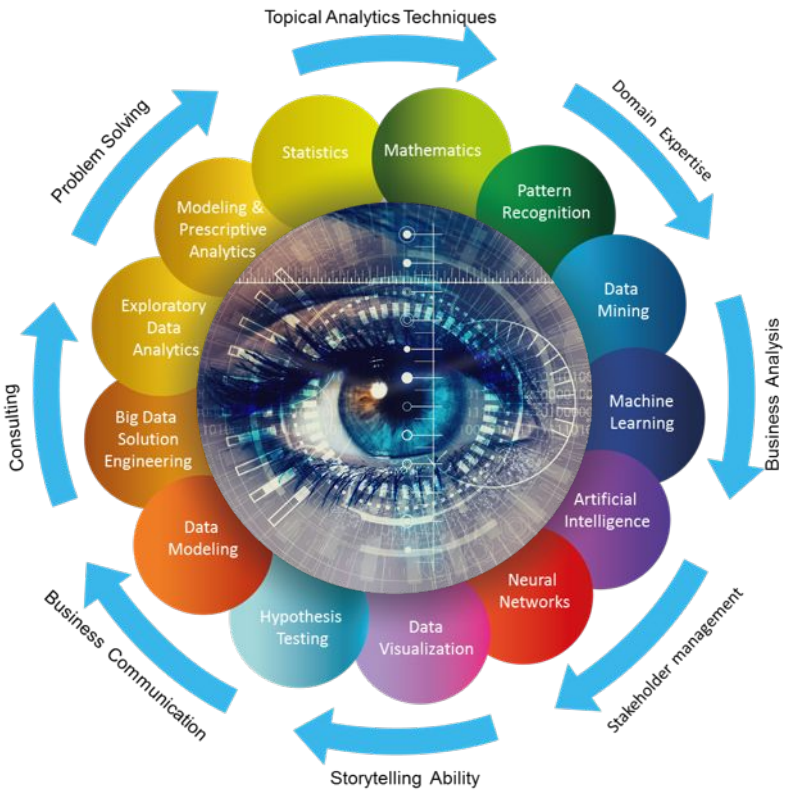
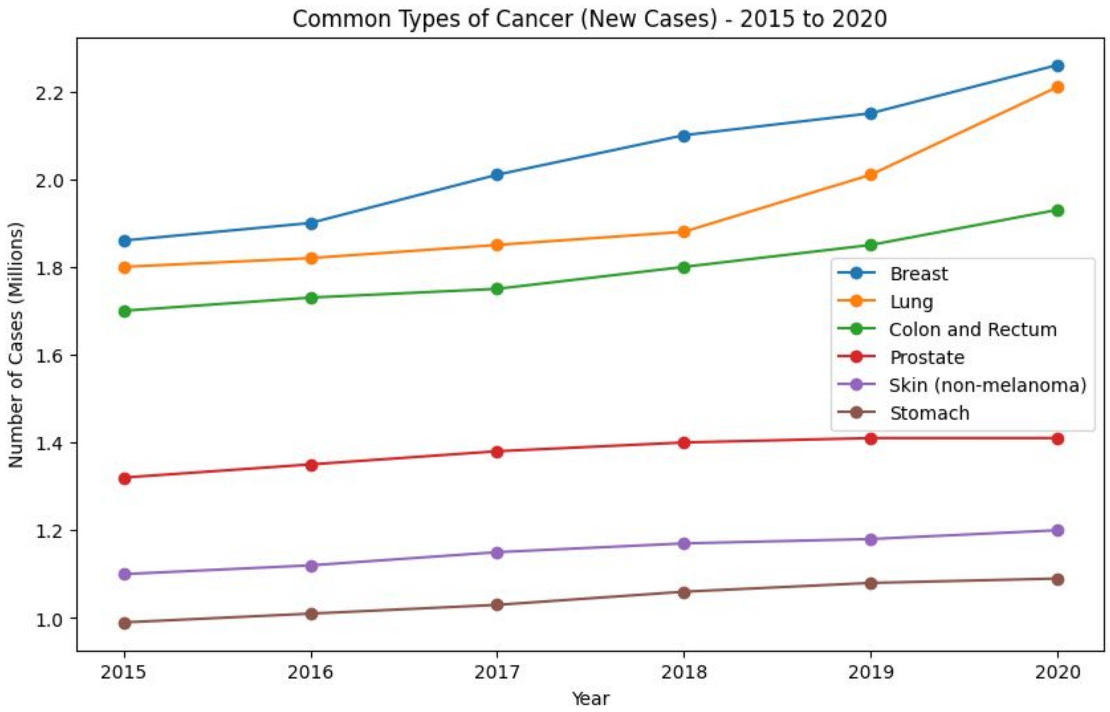


Introduction and aims



Motivation

- According to the WHO (World Health Organization), breast cancer is the most common in recent years
- There is useful information in unstructured notes in EHR (Electronic Health Records)
- Structuring this information can help, e.g. clinicians in their next clinical judgements

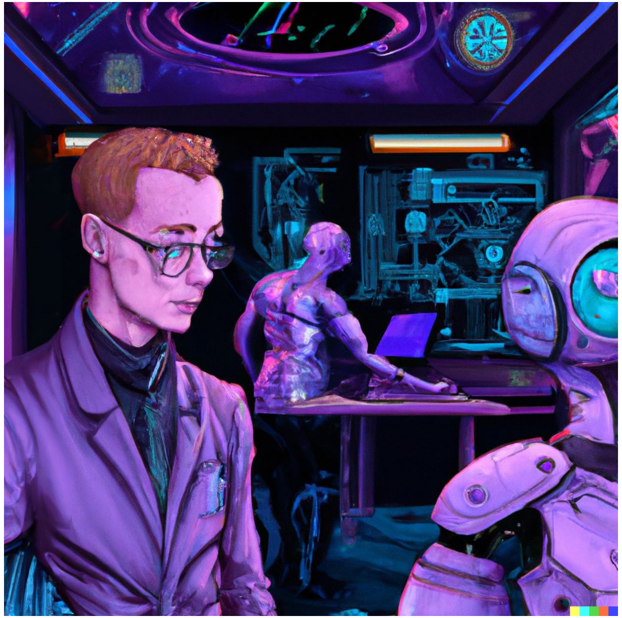


World Health Organization

Aims

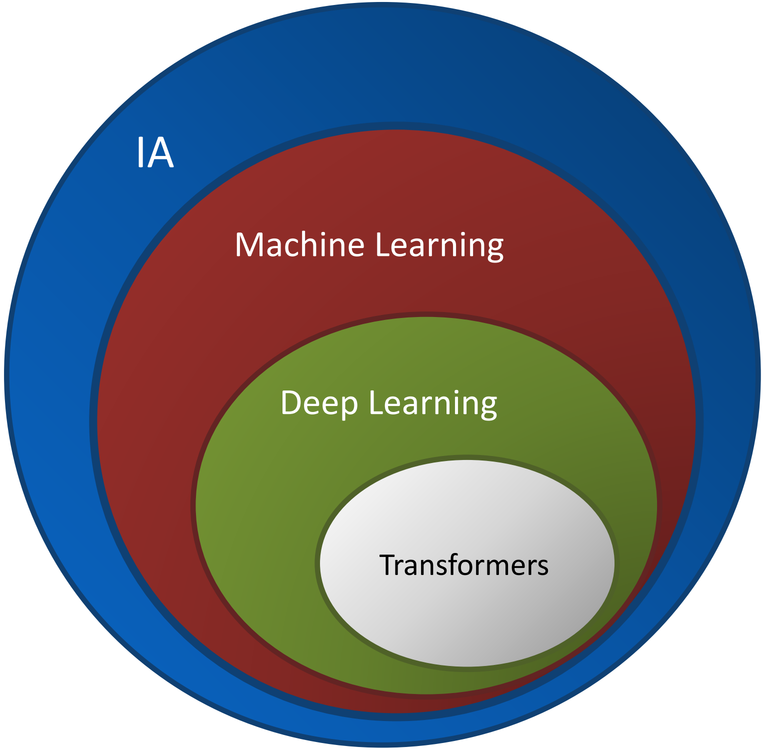
1. To evaluate NLP tools for extracting useful information from unstructured data in HCIS in Spanish
2. To convert the extracted information into structured data that can be used by computer systems in HCIS
3. Develop an automated NLP pipeline that can extract the most important information from a given clinical note

State of the Art

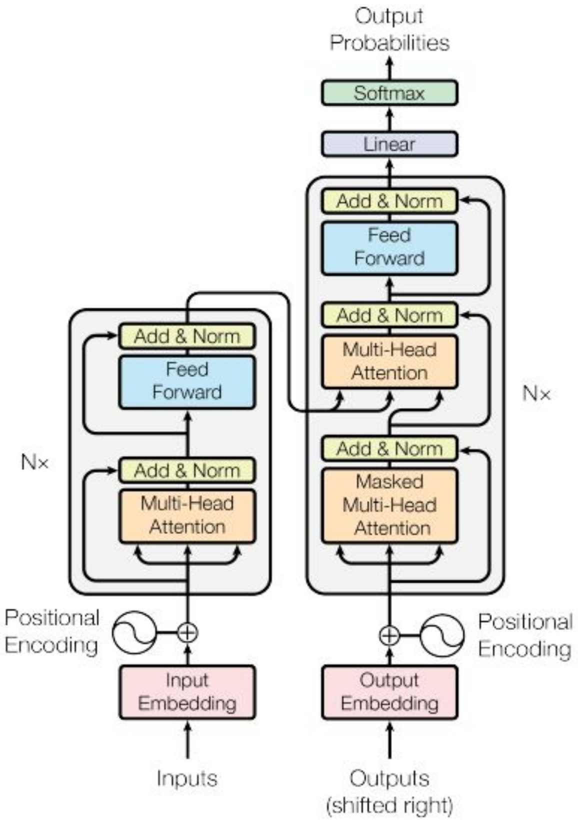


State of the art in NER

- Deep Learning is the most used technique for NLP today to extract entities from text (NER) Specifically, models with Transformer architecture are the ones that yield the best results (BERT)
- There is no corpus for NER focused on cancer in Spanish

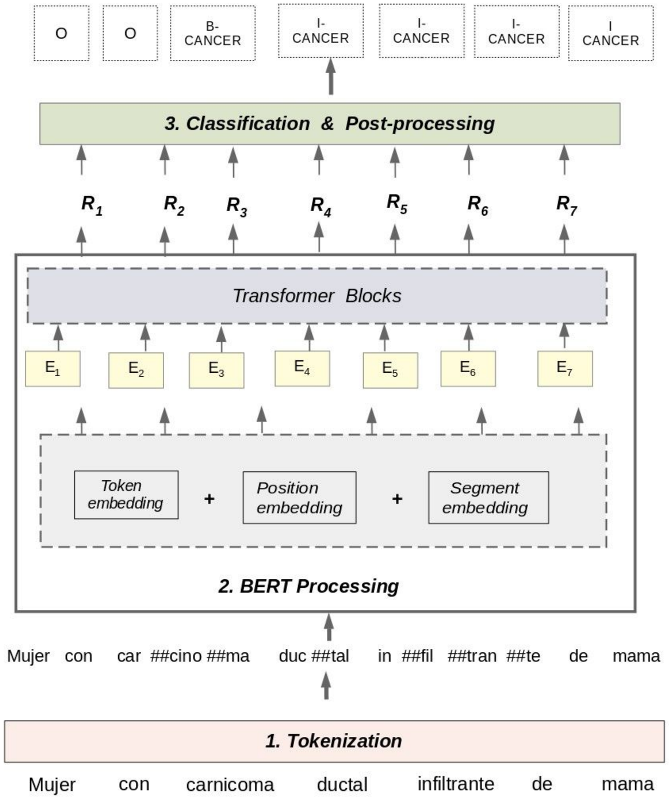


Transformers

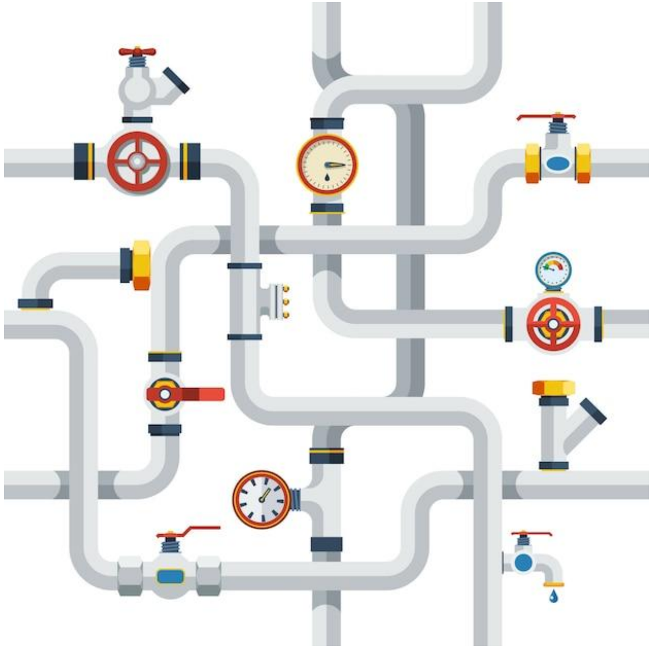


BERT (Bidirectional Encoder Representations from Transformers)

- Bidirectionally connected Transformers cells
  Encoder only
- Language Model (LM)
  Depending on the corpus with which it is pre-printed has a name, ours is called multilingual BERT
  It can be used for NER, doing a finetuning

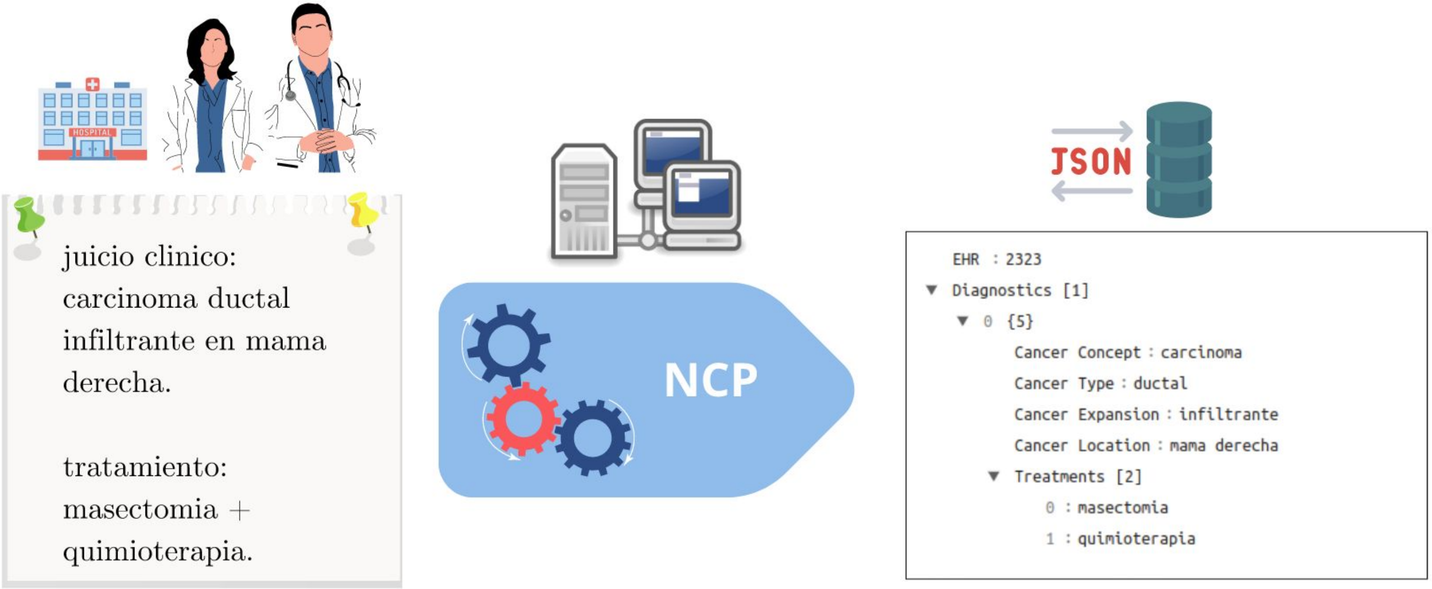


NCP - NLP Cancer Pipeline



Idea

- Module that converts a judgment note written by clinicians to JSON



NER (Named Entity Recognition)

- The initial step prior to structuring is to acquire the entities present in the text

evaluacion de la respuesta patologica (miller y payne):

g5 CANCER_GRADE ausencia de celularidad tumoral infiltrante CANCER_EXP en la mama CANCER_LOC y

tipo b ganglios linfaticos CANCER_LOC con metastasis CANCER_MET y

sin cambios por quimioterapia en valoracion ganglionar CANCER_LOC . no invasion CANCER_MET linfovascular CANCER_LOC

evaluacion de la respuesta patologica (miller y payne):

g5 ausencia de NEG celularidad tumoral infiltrante en la mama NSCO y

tipo b ganglios linfaticos con metastasis y

sin NEG cambios por quimioterapia en valoracion ganglionar NSCO . no NEG invasion linfovascular NSCO

Data Description

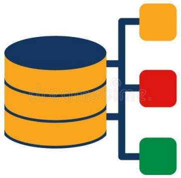
Data Source

- 3000 patient notes from clinical judgments
- Technical vocabulary with acronyms specific to each doctor
- Texts with grammatical errors

Manual Selection

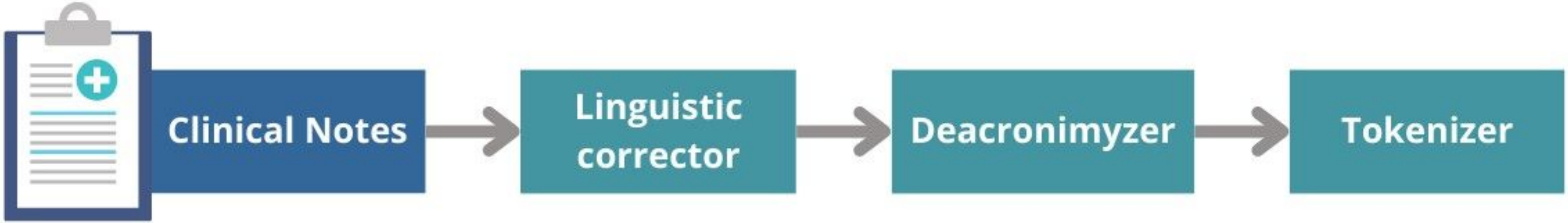
Corpus (without annotation)

550 notes from different years and different patients.



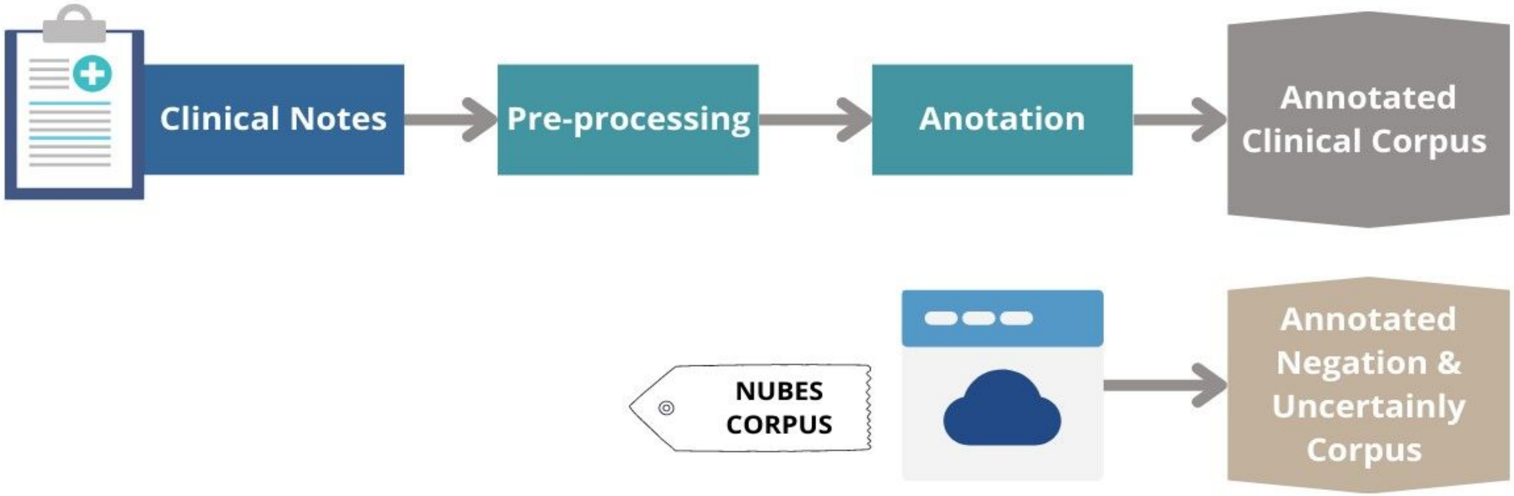
Preprocessing

- The purpose is to facility the learning of entities by the model Depending on the input format may consist of more phases as anonymization or removal of links
- It is applied before annotate and before the model predicts



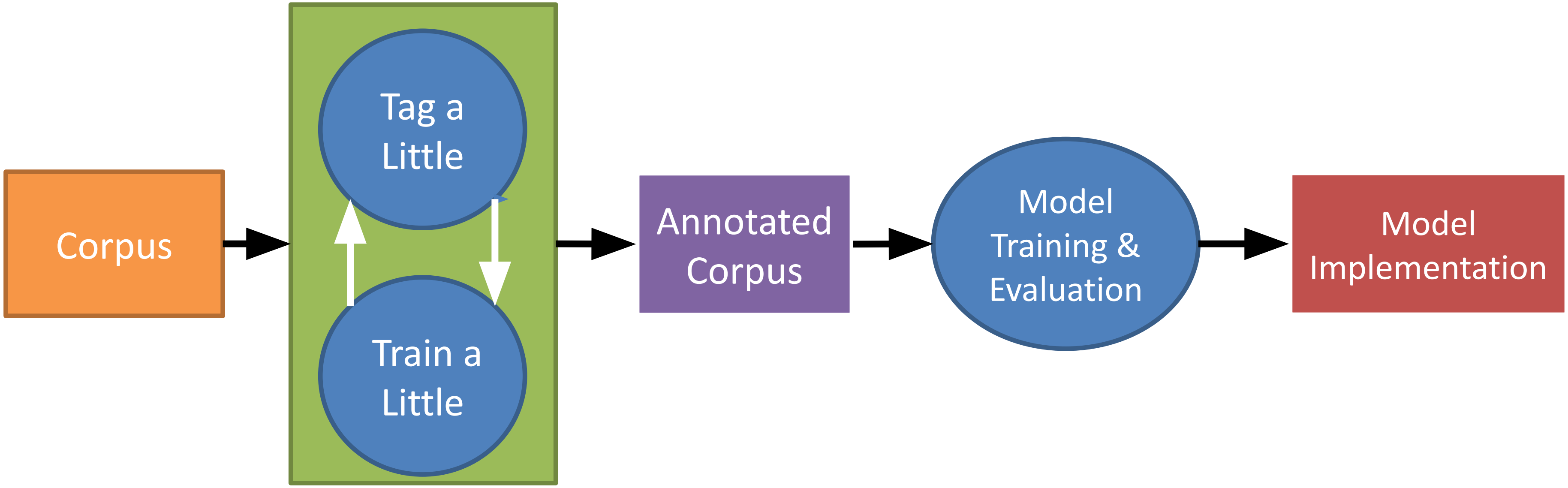
Corpus Collection

- Obtaining good data is crucial in NER



Stages to obtain a NER model

- Generating a NER model is an iterative and incremental process
  The number of entities required for each tag depends on the architecture

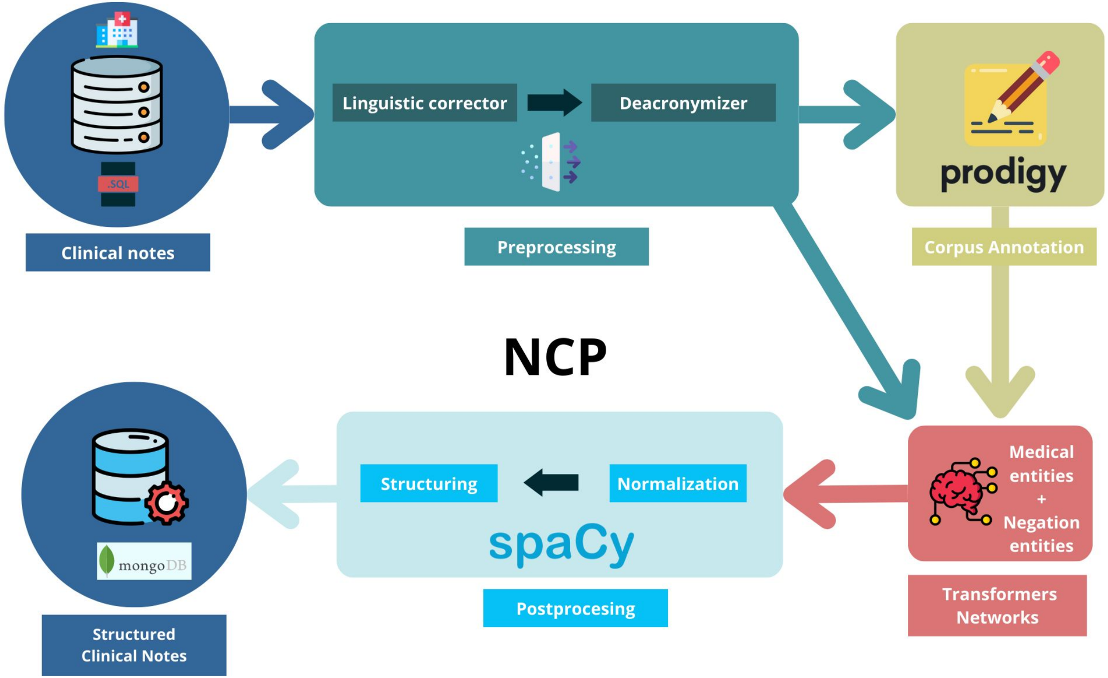


Corpus extraction

Tecnología

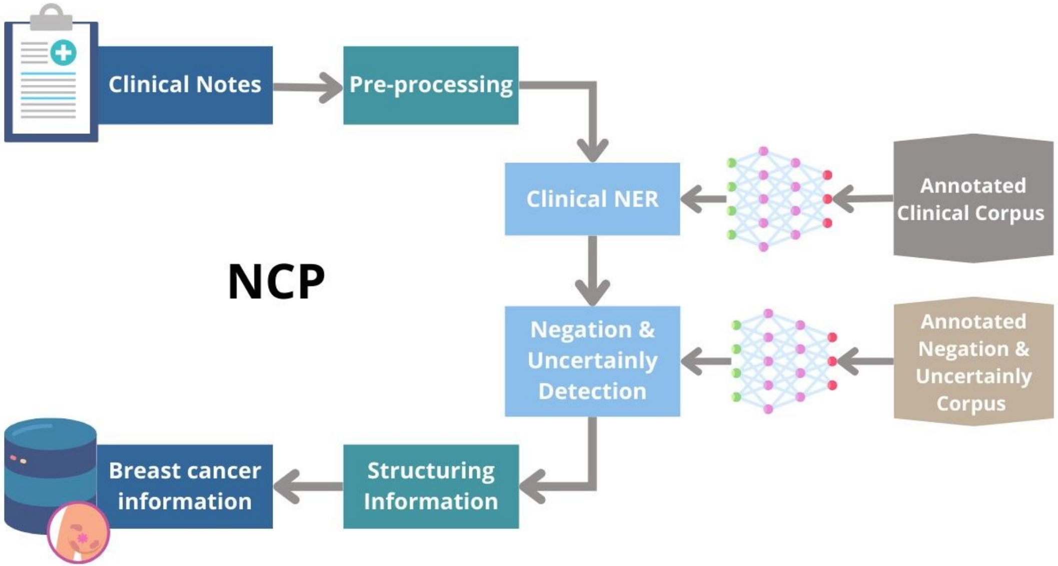
Software:

1. Leer las notas de Base de Datos SQL
2. Corregir y des-acronimizar las notas con regex.
3. Anotar las notas en Prodigy
4. Entrenar y evaluar modelos NER con spaCy
5. Usar los modelos de spaCy y técnica de similitud y regex para normalizar y estructurar.
6. Volcar las notas a MongoDB.



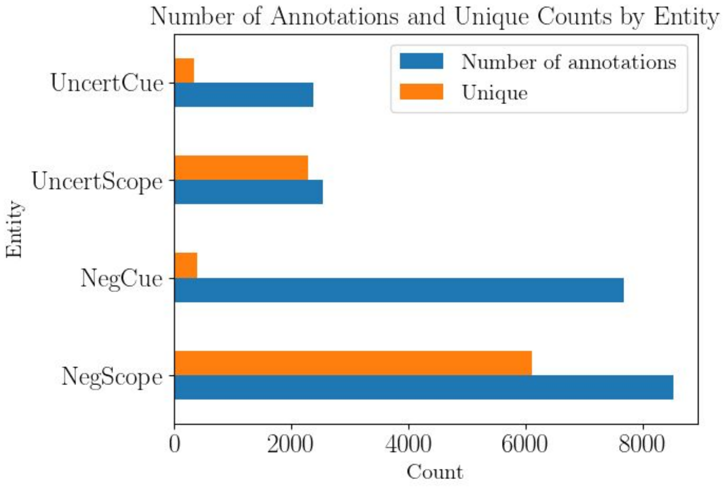
NLP Cancer Pipeline

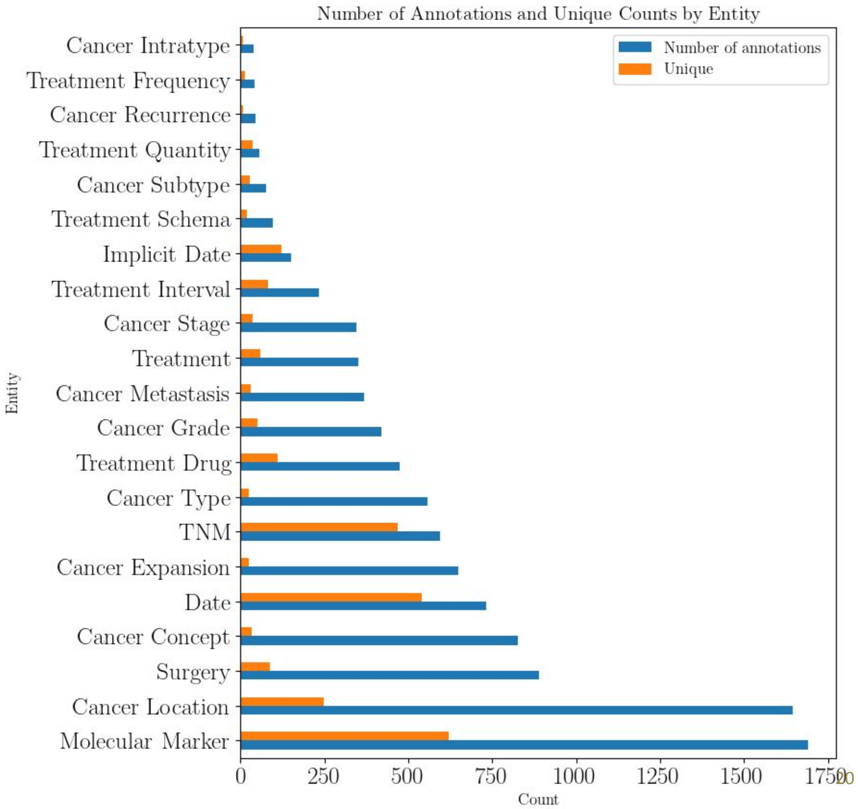
- Automated pipeline that processes notes in batches or online processing
- Create a structured cancer information format that can be easily analyzed or extracted



Experimentation

Distribución de las entidades





Magerit-3 CeSviMa

CeSViMa
CENTRO DE SUPERCOMPUTACIÓN Y VISUALIZACIÓN DE MADRID

Where do we run the tests?

Magerit-3: Efficient supercomputer

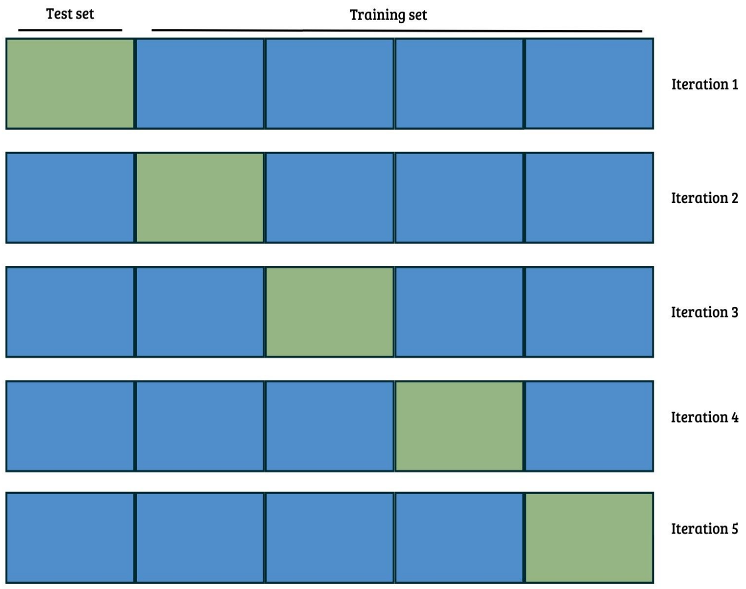
# Procesadores:	5.024 cores
Memoria:	49.920 GiB
Potencia (Rpeak)	
CPU:	392,60 TFLOPS
Aceleradores:	183,20 TFLOPS
Total:	575,80 TFLOPS

Cantidad	Procesador	RAM	HD	Aceleradores
72 nodos	2 × Intel® Xeon® Gold 6230 (20 cores @ 2.1 GHz)	192 GiB	480 GiB (SSD SATA)	—
48 nodos	2 × Intel® Xeon® Gold 6240R (24 cores @ 2.4 GHz)	768 GiB	120 GiB (SSD M.2)	—
4 nodos	2 × Intel® Xeon® Gold 6240R (24 cores @ 2.4 GHz)	192 GiB	128 GiB (SSD M.2)	4 × NVIDIA A100
2 nodos	2 × Intel® Xeon® Gold 6230 (20 cores @ 2.1 GHz)	192 GiB	128 GiB (SSD M.2)	2 × NVIDIA V100

Evaluate Clinical NER model

Feature / Model	NVIDIA RTX 3090	NVIDIA A100
Purpose	Gaming, general use	Specifically designed for artificial intelligence applications
Manufacturing Process Technology	8nm	7nm
Total core speed	1095 MHz	1395 MHz
Memory	24 GB GDDR6X	40 GB HBM2e
Price	1500 USD	10000 USD
Time in 10-fold	267m 61s ~ 4,45h	250m 52s ~ 4,17h

Fuente: https://askgeek.io/



* To take advantage of a supercomputer it is mandatory to develop parallel programs

Results & Discussion

- Clinical NER has good results
- Negation & Uncertainly NER has reasonable results
- Since obtaining clinical entities is conditioned on whether the sentence presents negation or uncertainty, it is necessary to assess the feasibility of including it in the pipeline

Model	Test	Precision	Recall	F-score
Clinical NER	10-fold	0.9346	0.9366	**0.9356**
Negation & Uncertainly NER	80-20	0.873	0.859	**0.866**

Conclusions and Future Work

Conclusions

NCP -NLP Cancer Pipeline-, capable of processing clinical notes in Spanish, producing a structured JSON file

Deep learning-centric methodology for structuring the extraction of breast cancer information

Necessity of detection of negation and uncertainty to struct clinical notes

Annotation schema proposed to extract detailed and specific concepts for cancer diagnosis and treatment

Machine learning models >> regex, but less interpretable

Future work

- Consider investigating alternative language models designed for processing Spanish clinical text, such as RoBERTa-bsc-bio-ehr
- Improve the performance of Negation & Uncertainly Model
- Pretrain a LLM with carcer domain literature to enhance the results
- Finetune LLMs text2text to struct more complex notes

Thanks!

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